

Problem sheet–2

1. Identity system/Invertibility

Consider a system S with input $x(t)$ and output $y(t)$. This system is obtained through a series interconnection of a system S_1 followed by a system S_2 . The input-output relationships for S_1 is

$$S_1 : y_1(t) = 2x_1(2t+2),$$

where $x_1(t)$ denote input signal.

- Determine S_2 if a cascade of S_1 and S_2 is an identity system.
- Do the same exercise for its discrete counterpart.
- Assume now that the system S_1 response is $y_1(t) = 2x_1(2t-2)$, under what conditions this system can be inversed? Does it conflict with the definition that a system is invertible if distinct inputs leads to distinct outputs?

2. Initial conditions being zero

- Show that if a system is either additive or homogeneous, it has the property that if the input is identically zero, then the output is also identically zero.
- Determine a system (either in continuous or discrete time) that is neither additive nor homogeneous but which has a zero output if the input is identically zero.
- From part (a), can you conclude that if the input to a linear system is zero between times t_1 and t_2 in continuous time or between times n_1 and n_1 in discrete time, then its output must also be zero between these same times? Explain your answer.

3. Periodicity of a time-invariant system

- Consider a time-invariant system with input $x(t)$ and output $y(t)$. Show that if $x(t)$ is periodic with period T , then so is $y(t)$. Show that the analogous result also holds in discrete time.
- Give an example of a time-invariant system and a non-periodic input signal $x(t)$ such that the corresponding output $y(t)$ is periodic.

4. Condition of initial rest

- Show that causality for a continuous-time linear system is equivalent to the following statement: For any time t_0 and any input $x(t)$ such that $x(t) = 0$ for $t < t_0$, the corresponding output $y(t)$ must also be zero for $t < t_0$. The analogous statement can be made for a discrete-time linear system.
- Find a nonlinear system that satisfies the foregoing condition but is not causal.
- Find a nonlinear system that is causal but does not satisfy the condition.
- Show that invertibility for a discrete-time linear system is equivalent to the following statement: The only input that produces $y[n] = 0$ for all n is $x[n] = 0$ for all n . The analogous statement is also true for a continuous-time linear system.
- Find a nonlinear system that satisfies the condition of part (d) but is not invertible.

5. Correlation

Correlation, a function which gives a measure of similarity of two signals when shifted by a certain (t) amount, for two signals, $x(t)$ and $h(t)$ is defined as

$$\varphi_{xh}(t) = \int_{-\infty}^{\infty} x(t + \tau)h(\tau)d\tau$$

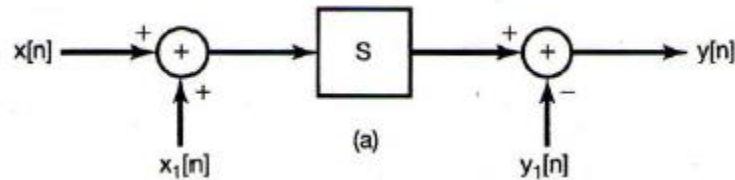
It is often important in practice to compute the correlation function $\phi_{hx}(t)$, where $h(t)$ is a fixed given signal, but where $x(t)$ may be any of a wide variety of signals. In this case, what is done is to design a system S with input $x(t)$ and output $\phi_{hx}(t)$.

- Is S linear? Is S time invariant? Is S causal? Explain your answers.
- Do any of your answers to part (a) change if we take as the output $\phi_{xh}(t)$ rather than $\phi_{hx}(t)$?

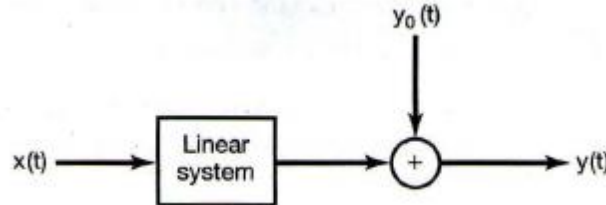
6. Incrementally linear systems

Let S denote an incrementally linear system, and let $x_1[n]$ be an arbitrary input signal to S with corresponding output $y_1[n]$. Consider the system illustrated in figure below.

- Show that this system is linear and that, in fact, the overall input-output relationship between $x[n]$ and $y[n]$ does not depend on the particular choice of $x_1[n]$.



- Use the result of part (a) to show that S can be represented in the form shown in figure below



- Which of the following systems are incrementally linear? Justify your answers and if a system is incrementally linear, identify the linear system L and the zero-input response $y_0[n]$ or $y_0(t)$ for the representation of the system as shown in the figure above

(i) $y[n] = n + x[n] + 2x[n + 4]$

(ii)
$$y[n] = \begin{cases} \frac{n}{2}, & n \text{ even} \\ \frac{n-1}{2} + \sum_{k=-\infty}^{\frac{n-1}{2}} x[k], & n \text{ odd} \end{cases}$$

7. System properties

In this chapter, we introduced a number of general properties of systems. In particular, a system may or may not be

- Memoryless
- Time invariant
- Linear
- Causal
- Stable

Determine which of these properties hold and which do not hold. Justify your answers, symbols have the usual meaning.

- $y(t) = x(t - 2) + x(2 - t)$
- $y(t) = [\cos(3t)]x(t)$
- $y(t) = \int_{-\infty}^{2t} x(\tau) d\tau$

$$d) \ y[n] = \sum_{k=-\infty}^{3n} x[k]$$

$$e) \ y(t) = \begin{cases} 0, & t < 0 \\ x(t) + x(t-2), & t > 0 \end{cases}$$

$$f) \ y(t) = \begin{cases} 0, & x(t) < 0 \\ x(t) + x(t-2), & x(t) > 0 \end{cases}$$

$$g) \ y(t) = x(t/3)$$

$$h) \ y(t) = \frac{d x(t)}{dt}$$

$$i) \ y(t) = t^2 x(t-1)$$

$$j) \ y[n] = x^2[n-2]$$

$$k) \ y[n] = x[n+1] - x[n-1]$$

$$l) \ y(t) = \text{Odd}\{x(t)\}$$

$$m) \ y(t) = x(\sin(t))$$

$$n) \ y[n] = \sum_{k=n-n_0}^{n+n_0} x[k] \text{ where } n_0 \text{ is a finite positive integer.}$$